DEWDRIVE · REGULATORY REGISTER

DewDrive regulatory and standards register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/07-regulatory.md @ 4f86e9e		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	G12: design gust 25 m/s (item 4, 2026-10-25)
0.1	2026-10-25	Amish Chadha	First register (kit 1.8 pipeline); sources checked 2026-10-25

DewDrive regulatory and standards register

DewDrive makes water that people may drink, handles a corrosive salt and carries a small lithium battery, so three sets of rules matter: drinking-water quality, materials in contact with the water, and the bought electrical parts. The open design itself is not placed on any market; rules on selling products apply to whoever builds and sells a unit. This register is not legal advice; anything marked "to assess" needs a qualified reviewer before any claim about the water or the product. Method: STANDARDS section 23. Sources checked 2026-10-25 where a link is given; standards named without a link were not opened in this session and are "to assess".

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	WHO *Guidelines for Drinking-Water Quality* and the WHO chloride fact sheet	International	Health-based guideline values; chloride has no health-based value, taste becomes detectable above about 250 mg/L	Reference for the collected water (PR-006); the device is not a treatment step and the documents require treatment before drinking	Partner NGO (water testing)	Applies as guidance; water test at TRL 4
G2	Directive (EU) 2020/2184 on water intended for human consumption, Annex I Part C	European Union	Indicator values: chloride 250 mg/L, aluminium 200 µg/L	Reference limits for chloride carry-over and aluminium from the condenser (PR-006); the national rule of the partner's country applies in practice	Partner NGO; local water authority	Applies as reference limit; national rule to assess
G3	Regulation (EC) No 1935/2004 (food-contact materials) and Regulation (EU) No 10/2011 (plastics)	European Union	Materials in contact with food and drinking water	Wetted parts: food-safe coating on the condenser top, gutter sealant, silicone drain tube, HDPE jerrycan (PR-006)	Design lead; supplier declarations	To assess: obtain supplier declarations of compliance
G4	NSF/ANSI/CAN 61, drinking water system components	United States and Canada	Health effects of materials in contact with drinking water	Same wetted parts as G3, for a North American build	Design lead	To assess
G5	GHS classification of calcium chloride (supplier safety data sheet)	International	Skin irritation category 2 (H315) and eye irritation category 2A (H319)	Preparing the sorbent and emptying the brine sumps: gloves and eye protection (build plan, safety stops)	Builder; user	Applies; met by the safety notes and PPE
G6	Food-additive grade calcium chloride (E 509 in the EU)	European Union (other countries have equivalents)	Purity of $CaCl_2$ sold for food use	The sorbent salt is bought food grade (BOM line 5), so trace impurities that could volatilise are limited	Builder	Applies by specification
G7	UN Manual of Tests and Criteria, section 38.3 (lithium battery transport)	International	Test summary required to ship lithium cells and batteries	The bought 12.8 V 6 Ah LiFePO4 battery (BOM line 12)	Supplier; builder	To assess: obtain the supplier's UN 38.3 test summary
G8	IEC 62133-2, safety of portable	International	Battery safety tests	The bought battery with its BMS	Supplier	To assess: supplier certificate

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	secondary lithium cells and batteries					
G9	Low Voltage Directive 2014/35/EU	European Union	Electrical equipment rated 50 to 1,000 V AC or 75 to 1,500 V DC	The device runs at 12.8 V DC	None	Does not apply
G10	Radio Equipment Directive 2014/53/EU (and FCC rules in the United States)	European Union; United States	Radio equipment placed on the market	The ESP32 logger has Wi-Fi and Bluetooth radio; a pre-certified module is specified	Builder who places a unit on the market	Does not apply to the open design; certification needed later for any sale
G11	IEC 61215 and IEC 61730 (PV module design and safety qualification)	International	PV module performance and safety	The bought 10 W panel (BOM line 11)	Supplier	To assess: supplier certificate
G12	EN 1991-1-4, wind actions on structures	Europe (national annexes)	Design wind speeds and force coefficients	The 25 m/s design gust (decided by Amish 2026-10-25, until the site value is known) and force coefficient in DWD-CAL-001 section H are first-order; the site wind speed comes from the national annex or local code	Design lead	To assess for the partner site
G13	EN ISO 13732-1, contact with hot surfaces	International	Burn thresholds by surface material and contact time	The 60 °C touch limit of PR-013 and the hot interior of the box	Design lead	To assess: confirm the threshold values against the standard

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- World Health Organization. *Chloride in Drinking-Water: Background Document for Development of WHO Guidelines for Drinking-Water Quality.* https://cdn.who.int/media/docs/default-source/wash-documents/wash-chemicals/chloride-fs-new.pdf (checked 2026-10-25).
- European Union. Directive (EU) 2020/2184 of the European Parliament and of the Council of 16 December 2020 on the quality of water intended for human consumption, Annex I. https://gesetze.legal/eu/rl_2020_2184_eu/anhang_i (checked 2026-10-25).

- Wikipedia. "Low Voltage Directive." https://en.wikipedia.org/wiki/Low_Voltage_Directive (checked 2026-10-25).
- Sigma-Aldrich. *Safety Data Sheet: Calcium Chloride Dihydrate*, version 4.5, 2014. https://isolab.ess.washington.edu//laboratory/safety/sds/calcium_chloride.pdf (checked 2026-10-25).
- Regulation (EC) No 1935/2004, Regulation (EU) No 10/2011, NSF/ANSI/CAN 61, UN Manual of Tests and Criteria section 38.3, IEC 62133-2, Directive 2014/53/EU, IEC 61215, IEC 61730, EN 1991-1-4 and EN ISO 13732-1: named by title; not opened in this session (to assess).